D4 Probability and statistics: problems

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Give probabilities to 3 decimals unless asked otherwise, as in a Dutch exam. Write down which GR (*grafische rekenmachine*) command you use and with which numbers; on an exam a bare answer earns few points. "*Bereken*" means calculate (exact or with the GR); "*toon aan*" means show that (give a full argument that ends in the given result); "*bewijs*" means prove.

Practice

1. A class has 7 girls and 5 boys. A committee of 4 pupils is chosen at random.

a. *Bereken* the probability that the committee has exactly 2 girls.
b. *Bereken* the probability that the committee contains at least one boy.

2. The word WISKUNDE has 8 different letters.

a. How many "words" of 4 different letters can be made from these letters?
b. How many different arrangements of all the letters of KANSBOOM are there?
c. In how many ways can you choose 4 of the 8 letters of WISKUNDE if order does not matter?

3. A bag contains 5 red and 3 blue marbles. You draw 3 marbles.

a. Without replacement (*zonder terugleggen*): *bereken* $P(\text{exactly 1 red})$ and $P(\text{at least 1 red})$.
b. With replacement (*met terugleggen*): *bereken* $P(\text{exactly 1 red})$ and $P(\text{at least 1 red})$.

4. In a survey of 200 pupils, 120 play a sport and 60 play an instrument; 36 do both.

a. Make a two-way table (*kruistabel*).
b. *Bereken* $P(\text{instrument} \mid \text{sport})$ for a randomly chosen pupil.
c. *Toon aan* that "plays a sport" and "plays an instrument" are independent for this group.

5. $P(A) = 0.4$ and $P(B) = 0.5$.

a. *Bereken* $P(A \cup B)$ if A and B are disjoint.
b. *Bereken* $P(A \cup B)$ if A and B are independent.
c. Can two disjoint events, both with positive probability, be independent? Explain.

6. At a school fair you pay €2 to throw two dice. If you throw double six you get €20. If you throw any other double you get €5. Otherwise you get nothing.

a. Make the probability distribution of the payout Y.
b. *Bereken* $E(Y)$ and the expected profit for the player per game.
c. *Bereken* $\sigma(Y)$.

7. A random variable X has the distribution below, and $E(X) = 1.6$.

x	0	1	2	3
$P(X = x)$	0.1	a	b	0.2

a. *Bereken* a and b.
b. *Bereken* $\sigma(X)$.

8. A packet of biscuits has mean weight 25 g and standard deviation 1.2 g. A box contains 16 packets; the weights are independent.

a. *Bereken* the mean and standard deviation of the total weight of the 16 packets.
b. *Bereken* the mean and standard deviation of the mean weight of the 16 packets.

c. The empty box itself weighs on average 30 g with standard deviation 2 g. *Toon aan* that the standard deviation of the weight of a full box is 5.2 g.
d. Assume all weights are normally distributed. *Bereken* the probability that a full box weighs less than 420 g.

9. A multiple-choice test has 20 questions, each with 4 options. Daan guesses every answer.

a. Explain why the number of correct answers X is binomially distributed.
b. *Bereken* $P(X = 5)$ and $P(X \geq 8)$.
c. *Bereken* $E(X)$ and $\sigma(X)$.

10. The heights of 15-year-old boys are normally distributed with $\mu = 172$ cm and $\sigma = 8$ cm.

a. *Bereken* the probability that a random boy is taller than 185 cm.
b. *Bereken* the percentage of boys between 160 and 180 cm.
c. *Bereken* the height that 90% of boys stay below.

11. A machine fills bottles with a normally distributed amount of liquid. The label says 990 ml.

a. The standard deviation is 4 ml. To what mean μ must the machine be set so that only 2% of bottles contain less than 990 ml?
b. Another machine has $\mu = 1000$ ml fixed. *Bereken* the largest σ for which at most 2% of bottles contain less than 990 ml.

Exam-level (*examenniveau*)

12. Screening. A rare disease affects 0.5% of the population. A test gives a positive result for 98% of ill people and for 3% of healthy people.

a. Make a two-way table for 100 000 people.
b. *Bereken* the probability that someone with a positive test is actually ill.
c. *Bereken* the probability that someone with a negative test is healthy.
d. Someone who tested positive is tested a second time, independently. The second test is also positive. *Bereken* the probability that this person is ill.

13. Quality control. A batch of 50 phone chargers contains 4 faulty ones. A buyer tests 5 chargers chosen at random (without replacement) and accepts the batch if none of them is faulty.

a. *Bereken* the probability that the batch is accepted.
b. *Bereken* the probability that exactly 1 tested charger is faulty.
c. A colleague computes (a) with the binomial distribution, $n = 5$, $p = 0.08$. What answer does she get? Explain why it is close to, but not the same as, the answer to (a).

14. Seeds. 85% of the seeds of a certain plant germinate, independently of each other. A packet contains 40 seeds.

a. *Bereken* the probability that at least 35 seeds germinate.
b. *Bereken* the expected number of seeds that germinate, and the standard deviation.
c. A gardener needs at least 30 plants. *Bereken* how many seeds a packet must contain so that the probability of at least 30 germinating seeds is at least 0.95.

15. Lift. A lift may carry at most 1000 kg. The weight of adults is normally distributed with $\mu = 80$ kg and $\sigma = 12$ kg.

a. 12 adults enter. *Bereken* the probability that the maximum load is exceeded.
b. *Bereken* the largest number of adults that may be allowed in so that the probability of exceeding 1000 kg is less than 1%.

16. Many dice. A fair die is thrown 600 times. X is the number of sixes.

a. *Bereken* $E(X)$ and $\sigma(X)$.

b. Use the normal approximation with continuity correction to estimate $P(X \leq 90)$.
c. Compare with the exact value from binomcdf, and with the value without continuity correction.
d. Estimate $P(95 \leq X \leq 105)$ with the normal approximation and compare with the exact answer.

17. A new medicine. An old medicine cures 60% of patients. A company claims its new medicine is better. In a trial with 25 patients, 20 are cured.

a. Formulate H_0 and H_1.
b. *Bereken* the p-value (*overschrijdingskans*) and draw a conclusion at $\alpha = 0.05$.
c. Determine the critical region (*kritieke gebied*).

18. A fair coin? To test whether a coin is fair, it is tossed 100 times. It lands heads 60 times.

a. Explain why this is a two-sided test, and formulate the hypotheses.
b. Test at significance level 5% using a p-value.
c. Determine the critical region.

19. Crisps. A manufacturer claims that bags of crisps contain on average 150 g, with $\sigma = 5$ g (normally distributed). A consumers' organisation suspects the bags contain too little and weighs 25 bags. The mean is 148.2 g.

a. Formulate the hypotheses and test at $\alpha = 0.05$.
b. Determine the critical region for the sample mean.

20. Study time. Six pupils recorded how many hours they studied (x) and their test score (y).

x	1	2	3	4	5	6
y	52	55	61	60	68	72

a. *Bereken* the correlation coefficient r and describe the relation in words.
b. *Bereken* the least-squares regression line.
c. Predict the score after 7 hours. Why would a prediction for 20 hours be meaningless?
d. Does this show that studying causes higher scores? Give an argument for and an argument against.

21. Waiting for a six. Sara throws a die until she gets a six, but she stops after at most 3 throws. X is the number of throws.

a. Make the probability distribution of X.
b. *Toon aan* that $E(X) = \frac{91}{36}$.
c. *Bereken* $\sigma(X)$.

Challenge

22. *Bewijs* (prove) for any random variable X with finitely many values:

a. $\text{Var}(X) = E(X^2) - \big(E(X)\big)^2$.
b. $\text{Var}(aX + b) = a^2\text{Var}(X)$.
c. $\text{Var}(X) \geq 0$, with equality only if X takes a single value with probability 1. Deduce that $E(X^2) \geq E(X)^2$.

23. Least squares in detail. For the regression line $y = ax+b$ with $a = S_{xy}/S_{xx}$ and $b = \bar{y}-a\bar{x}$:

a. Show that the residuals add up to 0.
b. Show that $\sum(x_i - \bar{x})e_i = 0$, where e_i is the residual of point i.

c. Show that the minimum value of Q is $S_{yy}(1-r^2)$, and deduce that $-1 \le r \le 1$.
d. Check (a) numerically with the data of problem 20.

24. Birthdays. Ignore leap years and assume all 365 birthdays are equally likely.

a. Show that the probability that k people all have different birthdays is $\prod_{i=0}^{k-1} \frac{365-i}{365}$.
b. Find the smallest k for which the probability of a shared birthday is more than 0.5.
c. Use $1-x \le e^{-x}$ to show that the probability of all different birthdays is at most $e^{-k(k-1)/730}$, and use this to explain why the answer to (b) is close to $\sqrt{730 \ln 2}$.

25. Letters and envelopes. n letters are put at random into n addressed envelopes, one per envelope. X is the number of letters in the correct envelope.

a. Use indicator variables to show that $E(X) = 1$ for every n.
b. For $n = 5$, count the arrangements in which no letter is correct, and *bereken* $P(X = 0)$. Compare with $1/e$.

26. Monty Hall. In a game show there are three doors; behind one is a car. You pick door 1. The host, who knows where the car is, always opens a different door with a goat behind it, choosing at random if he has two options, and offers you a switch. He opens door 3. Use Bayes' rule to *bereken* the probability that the car is behind door 2. Should you switch?

27. Errors of the second kind.

a. In problem 17 (critical region from part c), suppose the new medicine actually cures 80% of patients. *Bereken* the probability of a type II error.
b. In problem 19, suppose the true mean is 147 g. *Bereken* the probability that the test does **not** reject H_0.
c. Explain in words why the answer to (a) is so large, and what the company could do about it.

28. How big must a poll be?

a. *Toon aan* that $p(1-p) \le \frac{1}{4}$ for all p between 0 and 1.
b. A poll asks n people a yes/no question. The fraction of "yes" is $\hat{p} = X/n$, with X binomial. *Toon aan* that $\sigma(\hat{p}) = \sqrt{p(1-p)/n}$.
c. How large must n be to guarantee $\sigma(\hat{p}) \le 0.01$, whatever p is?